

Functional Enrichment Analysis of DEGs

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```
# Load all required packages
library(clusterProfiler)
library(enrichplot)
library(dplyr)
library(tidyr)
library(ggplot2)
library(pathview)
library(GO.db)
library(readr)
library(stringr)
library(UpSetR)
library(STRINGdb)
library(magick)

# TIFF output directory (created automatically)
tiff_dir <- "../results/TIFFs/4_FunctionalCategorization"
dir.create(tiff_dir, showWarnings = FALSE, recursive = TRUE)
```

1. Setup environment and directories

We create folders to organize our functional enrichment outputs.

```
out_dir <- "../results/enrichment"
kegg_dir <- "../results/enrichment/kegg"
adv_dir <- "../results/advanced_analysis"

dir.create(out_dir, showWarnings = FALSE, recursive = TRUE)
```

```
dir.create(kegg_dir, showWarnings = FALSE, recursive = TRUE)
dir.create(adv_dir, showWarnings = FALSE, recursive = TRUE)
```

2. Import and prepare data

2.1. Load functional metadata (eggNOG)

We load the eggNOG annotation and build the TERM2GENE objects to map genes to their functions.

```
# Load base annotation
eggnog_data <- read.delim("../data/annotation.tsv")

# GO Dictionary
term2gene_go <- eggnog_data %>%
  dplyr::select(GOs, X.query) %>%
  tidyr::separate_rows(GOs, sep = ",") %>%
  dplyr::filter(GOs != "" & GOs != "-" & !is.na(GOs)) %>%
  dplyr::mutate(query = trimws(X.query), GOs = trimws(GOs)) %>%
  dplyr::select(GOs, X.query) %>%
  as.data.frame()

# Map GO IDs to descriptive terms
vector_go <- Term(GOTERM)
term2name_go <- data.frame(
  GO_ID = as.character(names(vector_go)),
  Term = as.character(unname(vector_go)),
  stringsAsFactors = FALSE
)

# KEGG Dictionary (Filtered for 'map' metabolic pathways)
term2gene_kegg <- eggnog_data %>%
  dplyr::select(KEGG_Pathway, X.query) %>%
  tidyr::separate_rows(KEGG_Pathway, sep = ",") %>%
  dplyr::filter(KEGG_Pathway != "" & KEGG_Pathway != "-" & !is.na(KEGG_Pathway)) %>%
  dplyr::filter(grepl("map", KEGG_Pathway)) %>%
  dplyr::mutate(query = trimws(X.query), KEGG_Pathway = trimws(KEGG_Pathway)) %>%
  dplyr::select(KEGG_Pathway, X.query) %>%
  as.data.frame()
```

2.2. Import Differentially Expressed Genes (DEGs)

We extract and separate the significant genes found in our previous analysis.

```
# Import significant results and clean IDs
deg_table <- read_csv("../results/deg_Treatment_vs_Control.csv") %>%
  mutate(GeneID = str_extract(...1, "WP_[0-9]+\.[0-9]+"))

## New names:
## Rows: 216 Columns: 7
## -- Column specification
## ----- Delimiter: "," chr
```

```
## (1): ...1 dbl (6): baseMean, log2FoldChange, lfcSE, stat, pvalue, padj
## i Use `spec()` to retrieve the full column specification for this data. i
## Specify the column types or set `show_col_types = FALSE` to quiet this message.
## * `` -> `...1`
```

```
# Separate genes by regulation direction
up_genes <- deg_table %>% filter(log2FoldChange > 0) %>% pull(GeneID) %>% trimws()
down_genes <- deg_table %>% filter(log2FoldChange < 0) %>% pull(GeneID) %>% trimws()
all_deg_ids <- trimws(deg_table$GeneID)
```

3. Enrichment Analysis (ORA)

3.1. Gene Ontology (GO)

We test for over-represented biological processes in both up-regulated and down-regulated genes.

```
# ORA: Up-regulated
enrich_go_up <- enricher(gene = up_genes,
  TERM2GENE = term2gene_go,
  TERM2NAME = term2name_go,
  pAdjustMethod = "BH",
  pvalueCutoff = 0.1,
  qvalueCutoff = 0.1)

# ORA: Down-regulated
enrich_go_down <- enricher(gene = down_genes,
  TERM2GENE = term2gene_go,
  TERM2NAME = term2name_go,
  pAdjustMethod = "BH",
  pvalueCutoff = 0.1,
  qvalueCutoff = 0.1)
```

Create and export the Dotplots.

```
p_go_up <- enrichplot::dotplot(enrich_go_up, showCategory = 15,
  title = "GO Enrichment (Up-regulated)" +
  scale_y_discrete(labels = function(x) str_wrap(x, width = 45)) +
  scale_color_gradient(low = "#4575B4", high = "#D73027",
    name = "adj. p-value") +
  theme_minimal(base_size = 12) +
  theme(plot.title = element_text(face = "bold", size = 14),
    axis.text.y = element_text(size = 10),
    axis.text.x = element_text(size = 10),
    legend.title = element_text(size = 10))
```

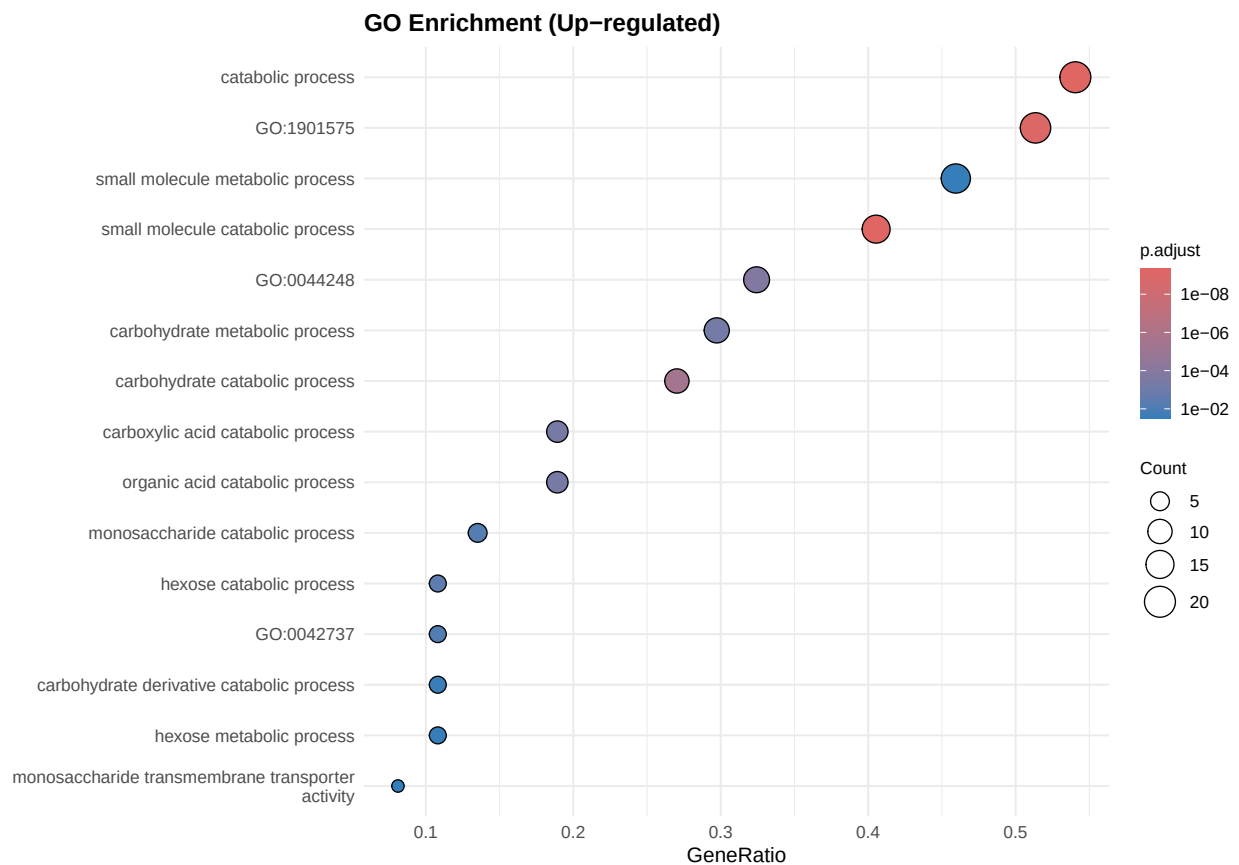
```
## Scale for y is already present.
## Adding another scale for y, which will replace the existing scale.
```

```
p_go_down <- enrichplot::dotplot(enrich_go_down, showCategory = 15,
                                title = "GO Enrichment (Down-regulated)" +
                                scale_y_discrete(labels = function(x) str_wrap(x, width = 45)) +
                                scale_color_gradient(low = "#4575B4", high = "#D73027",
                                name = "adj. p-value") +
                                theme_minimal(base_size = 12) +
                                theme(plot.title = element_text(face = "bold", size = 14),
                                axis.text.y = element_text(size = 10),
                                axis.text.x = element_text(size = 10),
                                legend.title = element_text(size = 10))
```

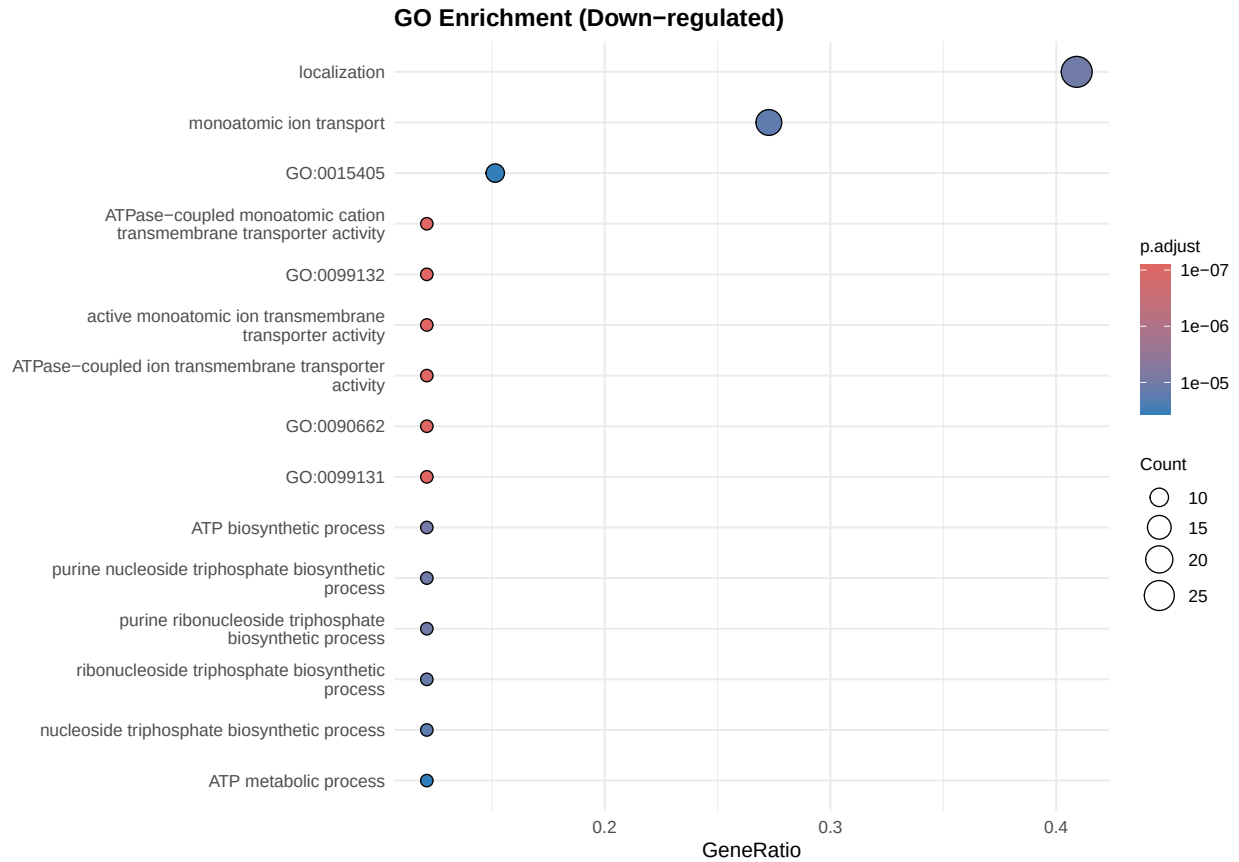
Scale for y is already present.

Adding another scale for y, which will replace the existing scale.

```
p_go_up
```



```
p_go_down
```



```
ggsave(filename = paste0(out_dir, "/dotplot_GO_UP.pdf"), plot = p_go_up, width = 12, height = 8)
ggsave(filename = file.path(tiff_dir, "dotplot_GO_UP.tiff"), plot = p_go_up, width = 12, height = 8)
ggsave(filename = paste0(out_dir, "/dotplot_GO_DOWN.pdf"), plot = p_go_down, width = 12, height = 8)
ggsave(filename = file.path(tiff_dir, "dotplot_GO_DOWN.tiff"), plot = p_go_down, width = 12, height = 8)
```

3.2. KEGG Pathway Enrichment

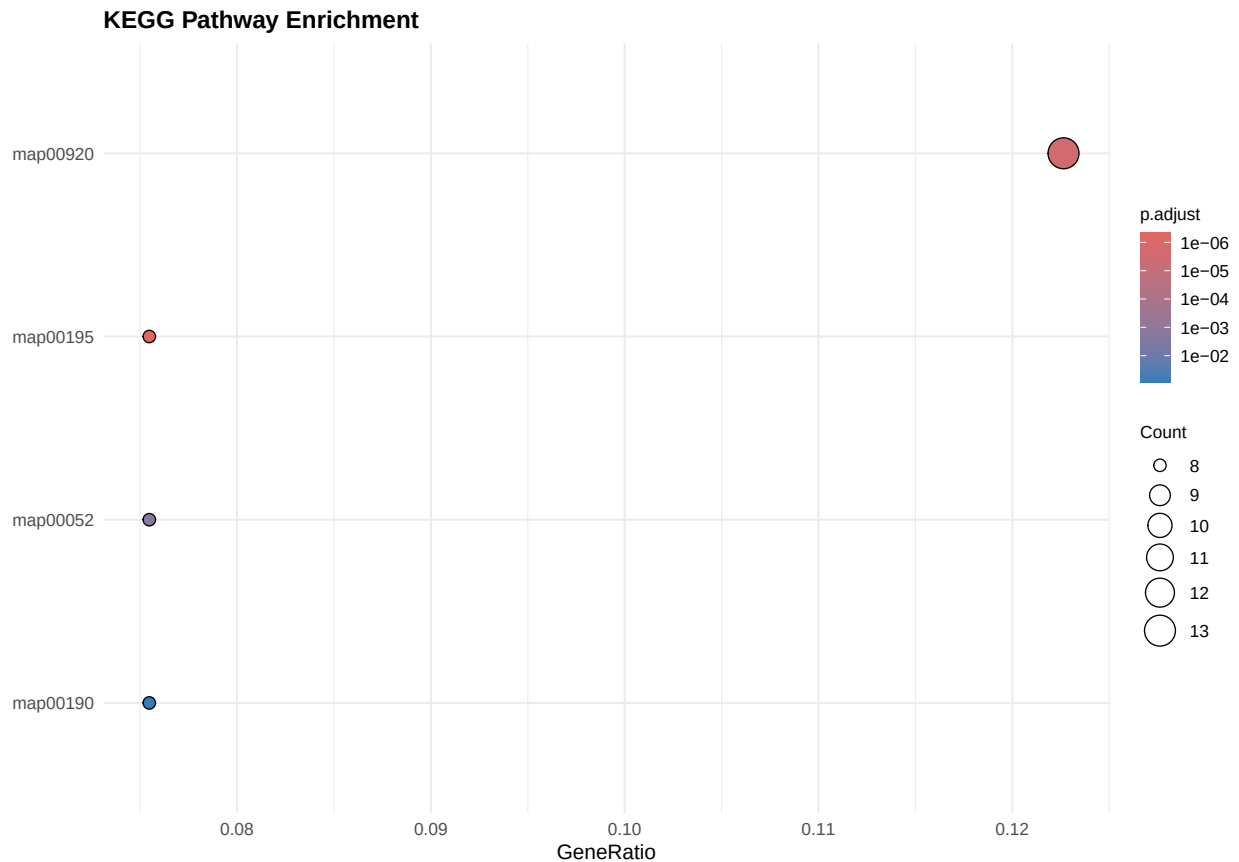
We check for affected metabolic pathways.

```
enrich_kegg <- enricher(gene = all_deg_ids,
  TERM2GENE = term2gene_kegg,
  pAdjustMethod = "BH",
  pvalueCutoff = 0.1,
  qvalueCutoff = 0.1)

p_kegg <- enrichplot::dotplot(enrich_kegg, showCategory = 15,
  title = "KEGG Pathway Enrichment") +
  scale_y_discrete(labels = function(x) str_wrap(x, width = 45)) +
  scale_color_gradient(low = "#4575B4", high = "#D73027",
    name = "adj. p-value") +
  theme_minimal(base_size = 12) +
  theme(plot.title = element_text(face = "bold", size = 14),
    axis.text.y = element_text(size = 10),
    axis.text.x = element_text(size = 10),
    legend.title = element_text(size = 10))
```

```
## Scale for y is already present.
## Adding another scale for y, which will replace the existing scale.
```

```
p_kegg
```



```
ggsave(filename = paste0(out_dir, "/dotplot_KEGG.pdf"), plot = p_kegg, width = 12, height = 8)
ggsave(filename = file.path(tiff_dir, "dotplot_KEGG.tiff"), plot = p_kegg, width = 12, height = 8, dpi
```

4. Automated KEGG Pathway Visualization (Pathview)

We download the top 15 affected metabolic pathways and color them using our `log2FoldChange` values.

```
# Get KEGG Orthology (KO) IDs
eggnog_ko <- eggnog_data %>%
  dplyr::select(X.query, KEGG_ko) %>%
  dplyr::filter(KEGG_ko != "" & KEGG_ko != "-") %>%
  dplyr::mutate(KEGG_ko = gsub("ko:", "", KEGG_ko)) %>%
  dplyr::mutate(KEGG_ko = sapply(strsplit(as.character(KEGG_ko), ","), `\[`, 1))
```

```

# Join with Fold Changes
pathview_data <- deg_table %>%
  dplyr::inner_join(eggnog_ko, by = c("GeneID" = "X.query")) %>%
  dplyr::filter(!is.na(KEGG_ko) & !is.na(log2FoldChange))

# Create a numeric vector for expression values
fc_vector <- pathview_data$log2FoldChange
names(fc_vector) <- pathview_data$KEGG_ko

```

Run the rendering loop.

```

# Extract top 15 KEGG IDs and clean the prefix
top_kegg_ids <- head(enrich_kegg$result$ID, 15)
top_kegg_clean <- gsub("map", "", top_kegg_ids)

# Temporarily change directory to save outputs neatly
current_wd <- getwd()
setwd(kegg_dir)

# 1. Loop to download and render each map (saved as PNGs)
for (path_id in top_kegg_clean) {
  message("Processing KEGG pathway: ", path_id)

  tryCatch({
    # Round limit to nearest integer so legend labels don't overlap
    fc_limit <- ceiling(max(abs(fc_vector)))
    pathview(gene.data = fc_vector,
              pathway.id = path_id,
              species = "ko",
              limit = list(gene = fc_limit, cpd = 1),
              kegg.native = TRUE)
  }, error = function(e) {
    message("Skipped pathway ", path_id, " (Insufficient data or API constraint).")
  })
}

# 2. Combine all maps into a single PDF safely
library(magick)

# Get all generated PNGs
png_files <- list.files(pattern = "\\pathview\\.png$")

# SAFETY FILTER: Keep only files larger than 5KB to avoid "ImproperImageHeader" errors
# from failed KEGG downloads (which are usually 0 bytes or tiny HTML error files).
valid_png_files <- png_files[file.info(png_files)$size > 5000]

if(length(valid_png_files) > 0) {
  message("Combining ", length(valid_png_files), " valid maps into a single PDF...")

  # Read only the valid, uncorrupted images
  kegg_maps <- image_read(valid_png_files)

  # Write to a single PDF

```

```

image_write(kegg_maps, format = "pdf", path = "Top15_KEGG_Pathways_Consolidated.pdf")

# Clean up all downloaded PNGs (both valid and corrupted) to keep the folder clean
file.remove(png_files)
} else {
  message("No valid maps were generated to combine.")
  # Clean up any leftover corrupted files
  file.remove(png_files)
}

# Return to original working directory
setwd(current_wd)

```

5. Network Analysis (PPI - STRINGdb)

We search for functional networks using orthologs confirmed by eggNOG.

```

# Significant genes (padj < 0.05) with a seed ortholog, capped at the 1001 lowest adjusted p-values
ppi_data <- deg_table %>%
  dplyr::inner_join(eggnog_data, by = c("GeneID" = "X.query")) %>%
  dplyr::filter(padj < 0.05 & seed_ortholog != "") %>%
  dplyr::arrange(padj) %>%
  dplyr::slice_head(n = 1001)

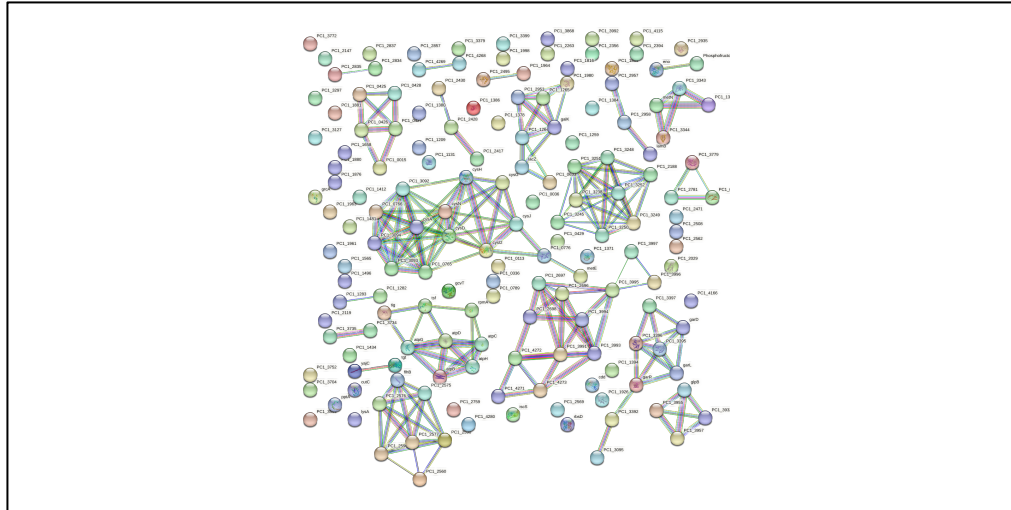
string_ids <- ppi_data$seed_ortholog
tax_id <- as.numeric(strsplit(string_ids[1], "\\.")[[1]][1])

string_db <- STRINGdb$new(version = "11.5",
  species = tax_id,
  score_threshold = 700,
  input_directory = out_dir)

string_db$plot_network(string_ids)

```

proteins: 209
interactions: 196
expected interactions: 58 (p-value: 0)



6. Specialized Plots

6.1. UpSet Plot (Intersections)

```
intersect_list <- list(
  "Up-regulated" = up_genes,
  "Down-regulated" = down_genes
)
```

```
p_upset <- upset(fromList(intersect_list),
  order.by = "freq",
  mainbar.y.label = "Intersection Size",
  sets.x.label = "Total Genes",
  text.scale = c(1.3, 1.3, 1, 1, 1.5, 1.2),
  point.size = 3.5,
  line.size = 1)
```

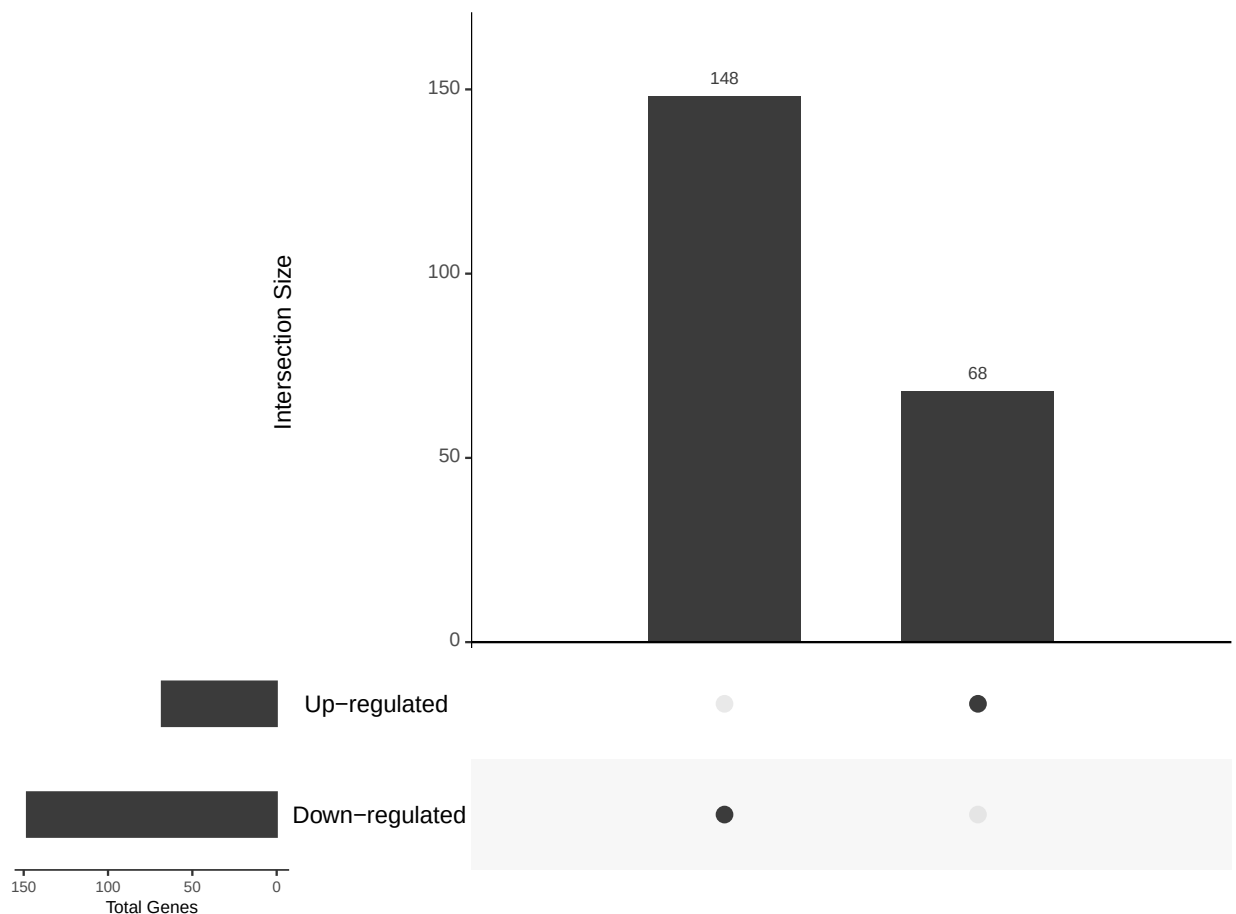
```
## Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
## i Please use tidy evaluation idioms with `aes()`.
## i See also `vignette("ggplot2-in-packages")` for more information.
## i The deprecated feature was likely used in the UpSetR package.
```

```
## Please report the issue at <https://github.com/hms-dbmi/UpSetR/issues>.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

## `geom_line()`: Each group consists of only one observation.
## i Do you need to adjust the group aesthetic?

## Warning: The `size` argument of `element_line()` is deprecated as of ggplot2 3.4.0.
## i Please use the `linewidth` argument instead.
## i The deprecated feature was likely used in the UpSetR package.
## Please report the issue at <https://github.com/hms-dbmi/UpSetR/issues>.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```
print(p_upset)
```



6.2. Enrichment Map

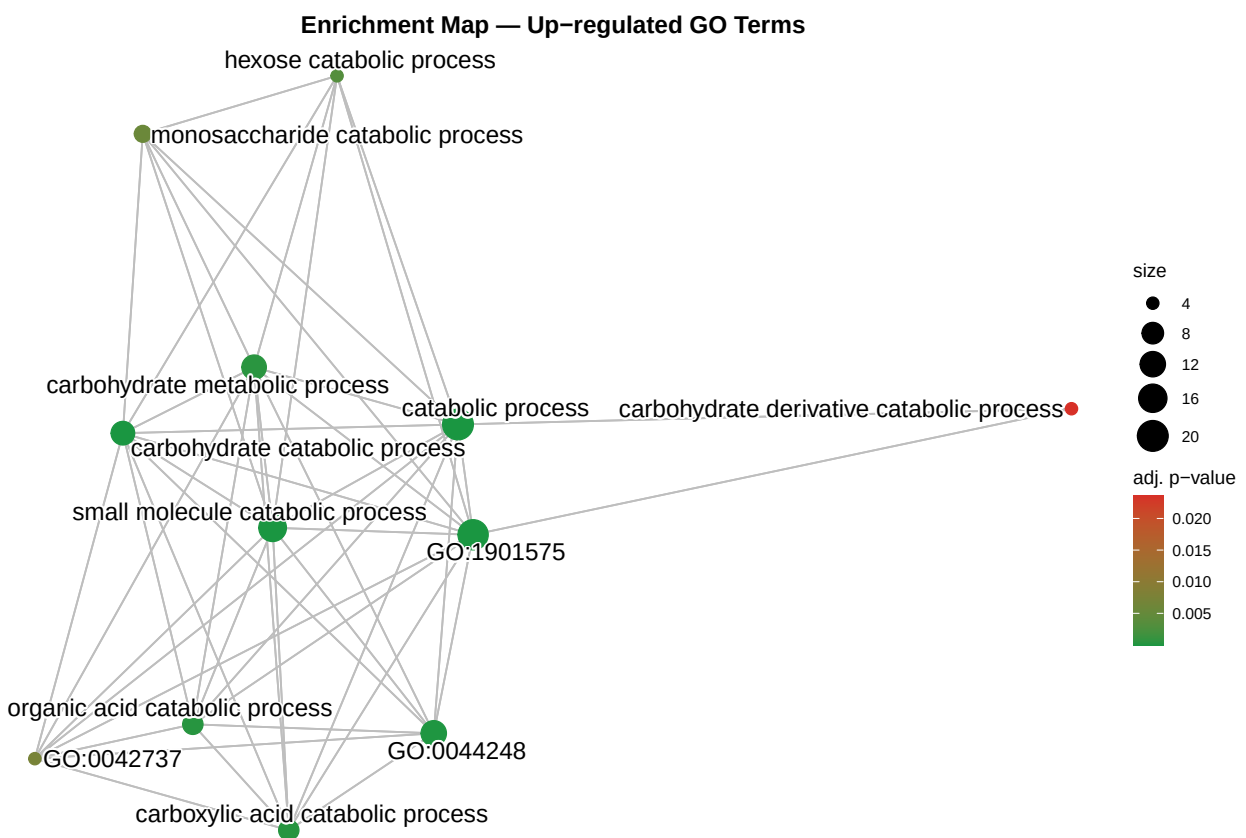
We use `emapplot` with a Fruchterman-Reingold layout (`fr`) which spreads similar terms more than `nicely`. Keeping only the top 12 terms avoids the central hairball typical when all enriched GO terms belong to the same super-family (e.g. purine/ATP biosynthesis).

```
# Up-regulated
enrich_go_up_sim <- pairwise_termsim(enrich_go_up)

p_emap_up <- emapplot(enrich_go_up_sim,
                      showCategory = 12,
                      color         = "p.adjust",
                      layout        = "fr") + # Fruchterman-Reingold spreads clusters better
scale_color_gradient(low = "#1A9641", high = "#D73027",
                    name = "adj. p-value") +
ggtitle("Enrichment Map - Up-regulated GO Terms") +
theme(plot.title = element_text(face = "bold", size = 14, hjust = 0.5),
      legend.text = element_text(size = 9))
```

```
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
```

```
p_emap_up
```



```

ggsave(filename = paste0(out_dir, "/emapplot_UP.pdf"),
        plot = p_emap_up, width = 11, height = 9)
ggsave(filename = file.path(tiff_dir, "emapplot_UP.tiff"),
        plot = p_emap_up, width = 11, height = 9, dpi = 300, compression = "lzw")

# Down-regulated
if (!is.null(enrich_go_down) && nrow(enrich_go_down@result) > 1) {
  enrich_go_down_sim <- pairwise_termsim(enrich_go_down)

  p_emap_down <- emapplot(enrich_go_down_sim,
                          showCategory = 12,
                          color         = "p.adjust",
                          layout        = "fr") +
    scale_color_gradient(low = "#1A9641", high = "#D73027",
                        name = "adj. p-value") +
    ggtitle("Enrichment Map - Down-regulated GO Terms") +
    theme(plot.title = element_text(face = "bold", size = 14, hjust = 0.5),
          legend.text = element_text(size = 9))

  p_emap_down
  ggsave(filename = paste0(out_dir, "/emapplot_DOWN.pdf"),
          plot = p_emap_down, width = 11, height = 9)
  ggsave(filename = file.path(tiff_dir, "emapplot_DOWN.tiff"),
          plot = p_emap_down, width = 11, height = 9, dpi = 300, compression = "lzw")
} else {
  message("Not enough down-regulated GO terms to build an enrichment map.")
}

```

```

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.

```

7. Prepare Data for Advanced Analysis (GSEA)

To run tests like Gene Set Enrichment Analysis (GSEA) or upstream regulator checks, we need the full dataset, not just the filtered significant genes.

```

# Import the full DESeq2 results (no p-value filter)
full_res <- read_csv("../results/deseq_results_Treatment_vs_Control.csv") %>%
  mutate(GeneID = str_extract(...1, "WP_[0-9]+\.[0-9]+")) %>%
  filter(!is.na(GeneID) & !is.na(log2FoldChange))

# Create a ranked list required for GSEA
ranked_list <- full_res %>%
  arrange(desc(log2FoldChange)) %>%
  dplyr::select(GeneID, log2FoldChange)

# Join the full data with eggNOG annotations
full_annotation <- full_res %>%
  left_join(eggnog_data, by = c("GeneID" = "X.query"))

```

```
# Save objects for the next pipeline
saveRDS(ranked_list, file = paste0(adv_dir, "/gsea_ranked_list.rds"))
write_csv(full_annotation, file = paste0(adv_dir, "/full_annotated_results.csv"))
```

Session information

```
sessionInfo()
```

```
## R version 4.5.2 (2025-10-31 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=Spanish_Mexico.utf8  LC_CTYPE=Spanish_Mexico.utf8
## [3] LC_MONETARY=Spanish_Mexico.utf8 LC_NUMERIC=C
## [5] LC_TIME=Spanish_Mexico.utf8
##
## time zone: America/Mexico_City
## tzcode source: internal
##
## attached base packages:
## [1] stats4      stats      graphics  grDevices  utils      datasets  methods
## [8] base
##
## other attached packages:
## [1] magick_2.9.1           STRINGdb_2.22.0
## [3] UpSetR_1.4.1           GO.db_3.22.0
## [5] AnnotationDbi_1.72.0   pathview_1.50.0
## [7] enrichplot_1.30.5      clusterProfiler_4.18.4
## [9] viridis_0.6.5          viridisLite_0.4.3
## [11] pheatmap_1.0.13        cowplot_1.2.0
## [13] lubridate_1.9.5         forcats_1.0.1
## [15] stringr_1.6.0          dplyr_1.2.0
## [17] purrr_1.2.1            readr_2.2.0
## [19] tidyr_1.3.2            tibble_3.3.1
## [21] tidyverse_2.0.0        EnhancedVolcano_1.28.2
## [23] ggrepel_0.9.7          ggplot2_4.0.3
## [25] DESeq2_1.50.2          SummarizedExperiment_1.40.0
## [27] Biobase_2.70.0         MatrixGenerics_1.22.0
## [29] matrixStats_1.5.0      GenomicRanges_1.62.1
## [31] Seqinfo_1.0.0          IRanges_2.44.0
## [33] S4Vectors_0.48.0       BiocGenerics_0.56.0
## [35] generics_0.1.4
##
## loaded via a namespace (and not attached):
## [1] splines_4.5.2          bitops_1.0-9           ggplotify_0.1.3
```

## [4] R.oo_1.27.1	polyclip_1.10-7	graph_1.88.1
## [7] XML_3.99-0.23	lifecycle_1.0.5	lattice_0.22-7
## [10] vroom_1.7.1	MASS_7.3-65	magrittr_2.0.4
## [13] rmarkdown_2.31	plotrix_3.8-14	yaml_2.3.12
## [16] otel_0.2.0	ggtangle_0.1.2	DBI_1.3.0
## [19] RColorBrewer_1.1-3	abind_1.4-8	R.utils_2.13.0
## [22] RCurl_1.98-1.19	hash_2.2.6.4	yulab.utils_0.2.4
## [25] tweenr_2.0.3	rappdirs_0.3.4	gdtools_0.5.0
## [28] tidytree_0.4.8	proto_1.0.0	codetools_0.2-20
## [31] DelayedArray_0.36.1	DOSE_4.4.0	ggforce_0.5.0
## [34] tidyselect_1.2.1	aplot_0.3.1	farver_2.1.2
## [37] jsonlite_2.0.0	systemfonts_1.3.2	chron_2.3-63
## [40] tools_4.5.2	ggnewscale_0.5.2	treeio_1.34.0
## [43] ragg_1.5.0	Rcpp_1.1.1	glue_1.8.0
## [46] gridExtra_2.3.1	SparseArray_1.10.10	xfun_0.55
## [49] qvalue_2.42.0	withr_3.0.3	fastmap_1.2.0
## [52] caTools_1.18.4	digest_0.6.39	timechange_0.4.0
## [55] R6_2.6.1	gridGraphics_0.5-1	textshaping_1.0.4
## [58] gtools_3.9.5	RSQLite_2.4.6	R.methodsS3_1.8.2
## [61] utf8_1.2.6	fontLiberation_0.1.0	data.table_1.18.2.1
## [64] httr_1.4.8	htmlwidgets_1.6.4	S4Arrays_1.10.1
## [67] scatterpie_0.2.6	sqldf_0.4-12	pkgconfig_2.0.3
## [70] gtable_0.3.6	blob_1.3.0	S7_0.2.1
## [73] XVector_0.50.0	htmltools_0.5.9	fontBitstreamVera_0.1.1
## [76] fgsea_1.36.2	scales_1.4.0	png_0.1-9
## [79] ggfun_0.2.1	knitr_1.51	tzdb_0.5.0
## [82] reshape2_1.4.5	curl_7.0.0	nlme_3.1-168
## [85] org.Hs.eg.db_3.22.0	cachem_1.1.0	KernSmooth_2.23-26
## [88] parallel_4.5.2	pillar_1.11.1	grid_4.5.2
## [91] vctrs_0.7.1	gplots_3.3.0	tidydr_0.0.6
## [94] cluster_2.1.8.1	Rgraphviz_2.54.0	evaluate_1.0.5
## [97] KEGGgraph_1.70.0	gsubfn_0.7	tinytex_0.60
## [100] cli_3.6.5	locfit_1.5-9.12	compiler_4.5.2
## [103] rlang_1.3.0	crayon_1.5.3	labeling_0.4.3
## [106] plyr_1.8.9	fs_2.1.0	ggiraph_0.9.6
## [109] stringi_1.8.7	BiocParallel_1.44.0	Biostrings_2.78.0
## [112] lazyeval_0.2.2	GOSemSim_2.36.0	fontquiver_0.2.1
## [115] Matrix_1.7-4	hms_1.1.4	patchwork_1.3.2
## [118] bit64_4.6.0-1	KEGGREST_1.50.0	igraph_2.2.2
## [121] memoise_2.0.1	ggtree_4.0.5	fastmatch_1.1-8
## [124] bit_4.6.0	ape_5.8-1	gson_0.2.0